

# Technology & Artificial Intelligence

*Foundations, applications, and responsible use*



## Overview

Modern AI systems combine **large datasets**, *scalable compute*, and refined algorithms. This document highlights terms with underline and **color** for emphasis. This section expands on the topic with a moderate amount of detail, weaving together context, rationale, and practical implications so that a parser encounters realistic prose of varying length.

■ **Definition:** A **transformer** is a neural architecture that relies on attention to weigh relationships between tokens.

## 1. Core Building Blocks

### Data, Models, Compute

The following paragraph is deliberately long. It stretches across several clauses and sentences, mixing subordinate structures, enumerations, and asides, so that downstream text-extraction and chunking behavior can be observed on a substantial block of running text. It continues past the point of a single line, wrapping naturally and giving the layout engine a meaningful column of words to flow. Longer paragraphs like this one help surface differences between loaders that split on paragraphs versus fixed token windows, and they exercise justification and leading in the rendered output as well.

- **Data:** curated, cleaned, representative
- **Models:** architecture + learned parameters
- **Compute:** accelerators for training and inference

#### *A Typical Pipeline*

- 1 Collect and label data
- 2 Train and validate a model
- 3 Evaluate against benchmarks
- 4 Deploy and monitor

## 2. Model Comparison

| Model Type  | Strength         | Typical Use   | Cost   |
|-------------|------------------|---------------|--------|
| CNN         | Spatial features | Vision        | Low    |
| RNN/LSTM    | Sequences        | Legacy NLP    | Medium |
| Transformer | Long context     | LLMs, vision  | High   |
| Diffusion   | Generation       | Images, audio | High   |

### Scaling Trends

| Year | Params (approx) | Trend              |
|------|-----------------|--------------------|
| 2018 | 340M            | Baseline           |
| 2020 | 175B            | Rapid growth       |
| 2023 | 1T+             | Mixture-of-experts |

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### Perspective

*“The real question is not whether machines think but whether men do.” — B. F. Skinner*

### 3. Responsible AI

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■ **Caution:** Models can amplify bias present in training data. Evaluate fairness before deployment.

#### Principles

- Transparency in capabilities and limits
- Human oversight for high-stakes decisions
- Privacy-preserving data handling

#### Looking Ahead

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## 4. Glossary & References

### Key Terms

| Term        | Meaning                              |
|-------------|--------------------------------------|
| Token       | A unit of text a model processes     |
| Inference   | Running a trained model on new input |
| Fine-tuning | Adapting a model to a narrower task  |
| Attention   | Weighing token relationships         |

### Further Reading

- 1 Foundational papers on attention
- 2 Surveys on responsible deployment
- 3 Benchmark documentation

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